

Solution Requirements

Date 03 October 2026
Team ID COMICCRAFT-TEAM
Project Name ComicCrafter AI – AI-Powered Comic Strip Generator
Maximum Marks 4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement	Category	Requirement Description	Priority
1	Story Prompt Input	Functional	Accept a short scenario from the user through Streamlit.	High
2	Art Style Selection	Functional	Allow Manga, Anime, American and Belgian styles.	High
3	Panel Generation	AI	Generate exactly six structured panel descriptions and dialogue using Gemini 1.5 Flash.	High
4	Image Generation	AI	Generate one image for each panel using ClipDrop text-to-image.	High
5	Comic Assembly	Processing	Add dialogue below each panel and combine six panels into a 3×2 grid.	High
6	PDF Export	Output	Create a PDF containing the final comic strip for download.	High
7	Validation & Errors	Reliability	Reject empty prompts, invalid styles and incomplete image generation with clear feedback.	High

Step 2: Non-Functional Requirements

- Usability: one-page Streamlit workflow with clear controls and progress messages.
- Maintainability: separate modules for panel generation, image generation and comic processing.
- Performance: generate six panels sequentially and report failures rather than silently producing an incomplete comic.
- Security: API keys must be stored in environment variables and not hard-coded.
- Portability: application should run in a Python environment with dependencies from requirements.txt.